Optimisation in my research

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Overview

- My research has used optimisation in a number of ways:
  - **Distributed optimisation and control algorithms for power systems (main focus today)**

  - Linear Matrix Inequalities (LMIs) for control synthesis*
  - Optimal power-flow in *unbalanced* low-voltage distribution networks*

(*slides at the end if anyone is interested)

Distributed optimisation / control

- Main idea:
  - each agent / device / system controls its own variables (typically state variables of that system which are also design variables of an optimisation problem)
  - by low band-width, distributed communication with other local agents, these variables are updated until the global optimisation problem is solved.
  - this can allow large-scale optimisation problems to be solved automatically by simple controllers at each device as opposed to solving the problem centrally at high computational cost and broadcasting the solution to each device.
  - already used (wide range of applications) in industry, although not especially common.
  - potential applications (mostly in the future) include coordination of UAVs, self-driving vehicle platoons, smart grid, etc.

Distributed optimization / control

- Research areas:
  - Improving convergence rates and properties (e.g. ADMM is one of the best recent algorithms)
  - Coping with enforced dynamics in each agent. Agents typically have their own application-specific dynamics which may not be easy to control.
  - Cyber-security – what happens if one or more devices are adversely controlled or deliberating send false data?
  - Coping with time delays and/or noise in communication
  - Stochastic networks – i.e. random communication failures, etc. – what can the distributed
  - Hybrid approaches using some form of reinforcement learning as well.

Application to power systems

- replace optimal power-flow and frequency control with distributed optimisation & control
- each generator has a controller that guides its power setpoints to the (true) OPF setpoint, based on communication with neighbours.
- not really needed in the power system of today, but potential for implementation in future grids / microgrids

Intro to distributed optimisation / control

- Consider a separable cost function for a network of agents

$$\min_{x \in \mathbb{R}^{|\mathcal{V}|}} \quad \sum_{i \in \mathcal{V}} F_i(x_i)$$

i = agent i
x_i = state of agent i
V = set of agents
$F_i(x_i)$ = cost function of agent i

- Constraint: at steady-state, each agent must have the same value of the state variable (this is called a *consensus* problem). Typically this is something like a market price or (scaled) power output, etc.

$$\begin{aligned} \min_{x \in \mathbb{R}^{|\mathcal{V}|}} \quad & \sum_{i \in \mathcal{V}} F_i(x_i), \\ \text{s.t.} \quad & \mathcal{A}^T x = 0, \end{aligned}$$

Communication network matrix

Intro to distributed optimisation / control

- KKT conditions for optimality

$$\nabla F(x^\star) = -\mathcal{A}\mu^\star,$$
$$\mathcal{A}^T x^\star = 0,$$

where $F(x) = \sum_{i\in\mathcal{V}} F_i(x_i)$ and $\mu^\star := [\mu_j^\star]_{j\in\mathcal{E}} \in \mathbb{R}^{|\mathcal{E}|}$ is the optimal value for the dual variable $\mu := [\mu_j]_{j\in\mathcal{E}} \in \mathbb{R}^{|\mathcal{E}|}$. For strictly convex cost functions, this problem can be solved in a distributed fashion using the *primal-dual* algorithm

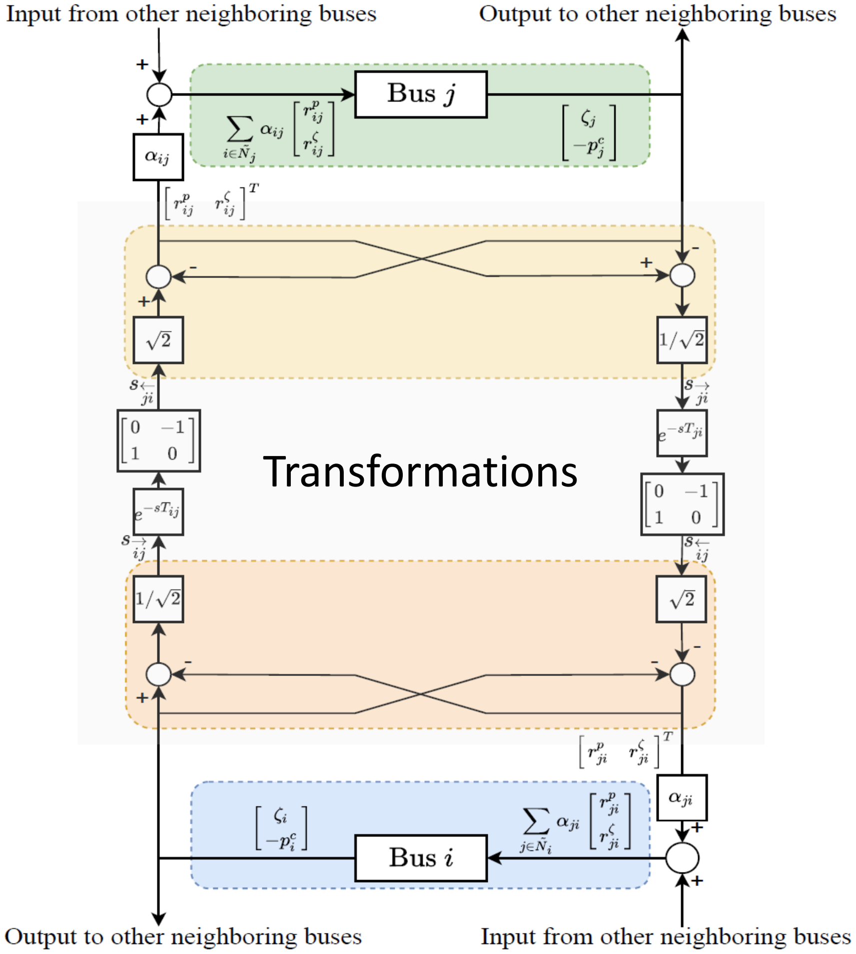
$$\dot{x} = -\nabla F(x) - \mathcal{A}\mu,$$
$$\dot{\mu} = \mathcal{A}^T x.$$

Guaranteed stability & optimality!!

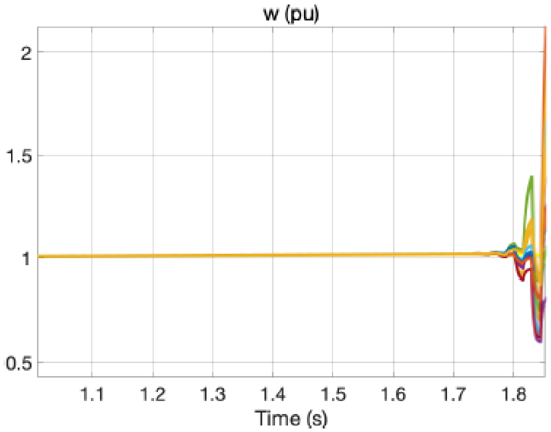
Results also exist for cost functions which are not strictly convex, but not in all cases and not with the same simplicity

Primal-dual secondary frequency control with delays (IEEE Trans. Automatic Control 2024)

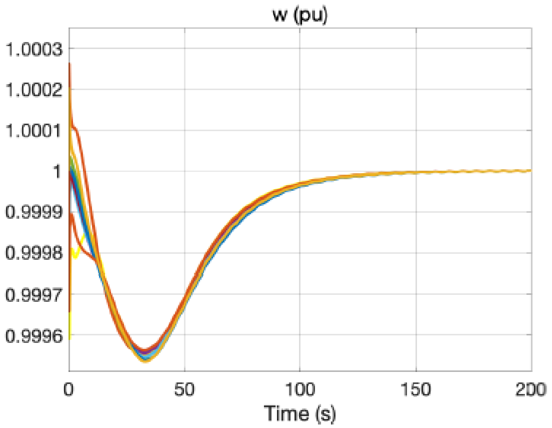
Idea: transform the communicated variables to allow noise resistance + delay robustness



Results (IEEE 39 system) with 20ms communication delays

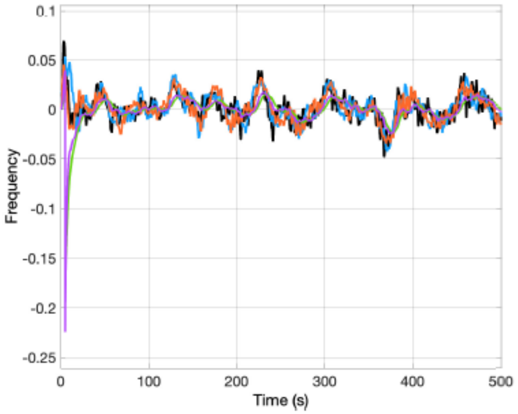


Traditional controller

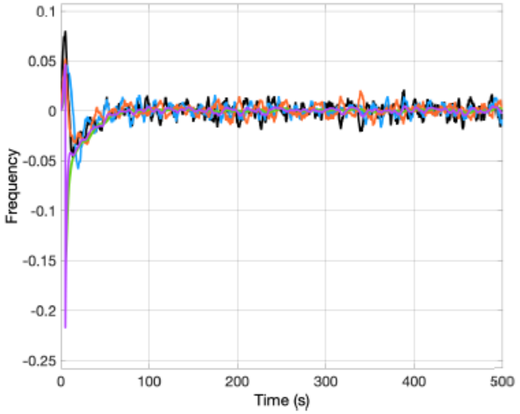


Our proposed controller

Results (IEEE 39 system) with noisy measurements (SNR = 20dB)



Traditional controller



Our proposed controller

Distributed secondary control for hybrid AC/DC grids (IEEE Trans. Control Systems 2021)

Idea: implement a similar form of distributed optimisation on hybrid AC/DC power grids

Optimisation formulation

$$\min_{p^G} C_G = \frac{1}{2}(p^G)^T Q p^G$$

$$\text{s.t.: } \mathbf{1}^T p^G = \mathbf{1}^T p^L + \mathbf{1}^T \begin{bmatrix} D\omega^G \\ 0 \\ 0 \end{bmatrix}$$

$$p_j^G = 0, \quad j \in X_{\mathrm{ac}}$$

System dynamics

$$\dot{\eta}_{ij} = \omega_i - \omega_j, \quad (i, j) \in E_{\mathrm{ac}} \tag{1a}$$

$$M_j \dot{\omega}_j = p_j^G - p_j^L + \sum_{i:i\to j} p_{ij} - \sum_{k:j\to k} p_{jk} - D_j \omega_j \qquad j \in N_{\mathrm{ac}} \setminus X_{\mathrm{ac}} \tag{1b}$$

$$0 = p_j^G - p_j^L + \sum_{i:i\to j} p_{ij} - \sum_{k:j\to k} p_{jk} - p_j^X, \quad j \in X_{\mathrm{ac}} \tag{1c}$$

$$C_j \dot{V}_j = p_j^G - p_j^L + \sum_{i:i\to j} p_{ij} - \sum_{k:j\to k} p_{jk} - p_j^X, \quad j \in N_{\mathrm{dc}} \tag{1d}$$

$$p_{ij} = \begin{cases} B_{ij} \sin \eta_{ij}, & j \in N_{\mathrm{ac}} \\ G_{ij}(V_i - V_j), & j \in N_{\mathrm{dc}}, \end{cases} \quad (i, j) \in E. \tag{1e}$$

Distributed Averaging Proportional-Integral Controller

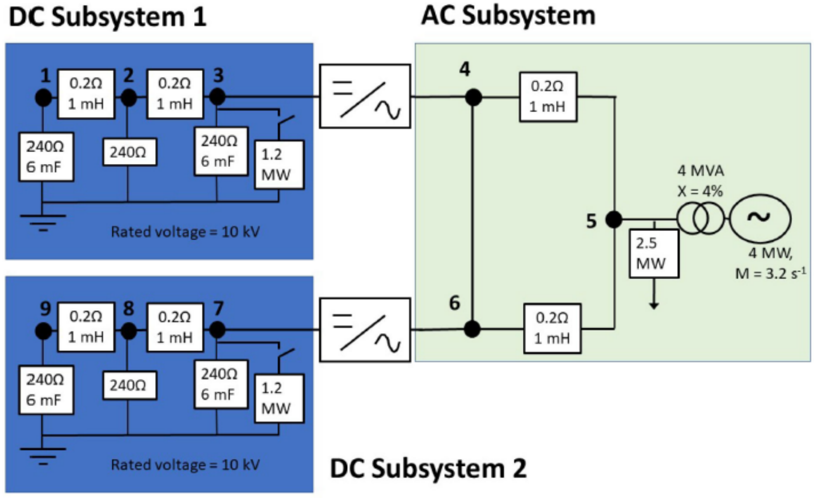
$$T_\xi \dot{\xi} = -\mathcal{L}\xi - \tilde{Q}\hat{\omega}$$

$$p^G = \tilde{Q}\xi - \mathcal{K}\hat{\omega}$$

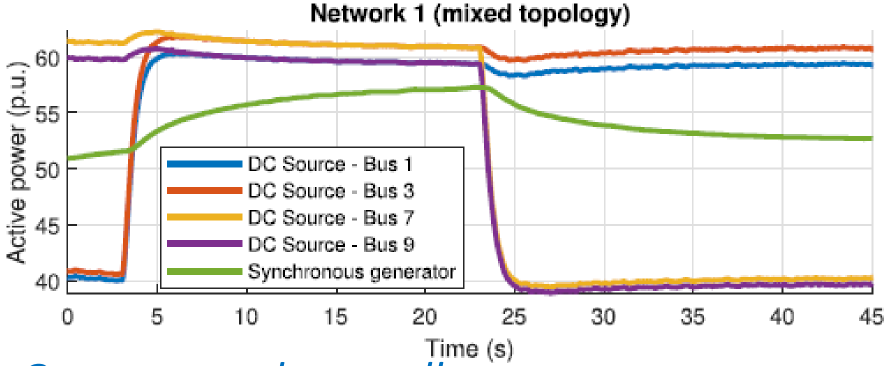
Control + Optimisation theory used to prove stability and optimality

Theorem 4 (Convergence to Optimality): Consider the dynamical system described in (1) and (2) with the control scheme (12), (13), and (15) and an equilibrium point for which Assumption 2 is satisfied. Then, there exists an open neighborhood of this equilibrium point such that all solutions of (1), (2), (12), (13), and (15) starting in this region converge to a set of equilibria that solve the optimization problem (4), with $\omega^* = 0$ and $\bar{V}^* = 0$.

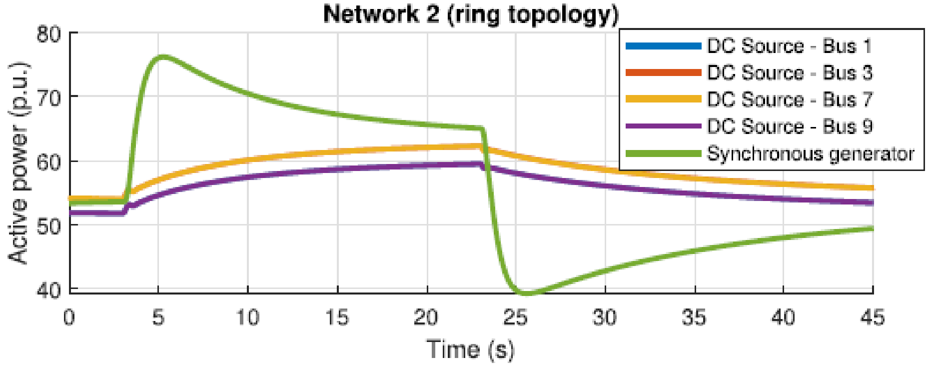
Case study



Traditional controller



Our proposed controller



Linear Matrix Inequalities

- State-of-the-art linear control techniques are usually based on Linear Matrix Inequalities (LMIs), where the control gains are the solution to a linear optimisation problem. Examples:
  - LQR (Riccati equations are LMIs)
  - Model predictive control
  - H-infinity, H2 control
  - Passivity-based control
  - Stability analysis
- Further reading (if interested): https://web.stanford.edu/~boyd/lmibook/lmibook.pdf

Scalable microgrid control (IEEE Trans. Smart Grid 2021)

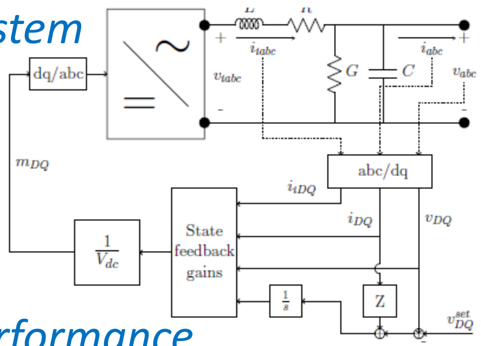
Formulation

Theorem 2: Consider the state feedback controller $u = -K\tilde{x} - Mw$ in conjunction with the dynamics (6). The system with input w and output z satisfies the passivity condition in Assumption 2 and is output-strictly passive with strictness index ρ if there exists a symmetric matrix $P = P^T > 0$ that satisfies:

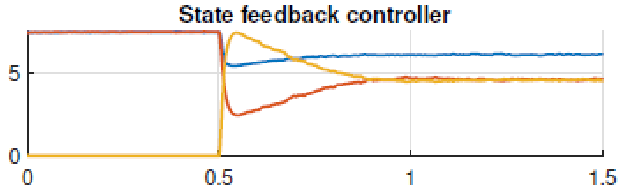
$$\begin{bmatrix} A_c^T P + PA_c + 2\rho C_c^T C_c & PB_c - C_c^T + 2\rho C_c^T D_c \\ B_c^T P - C_c + 2\rho D_c^T C_c & 2\rho D_c^T D_c - D_c^T - D_c \end{bmatrix} \le 0 \tag{10}$$

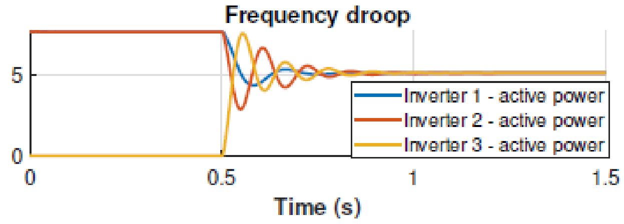
where $A_c = \tilde{A} - \tilde{B}_u K$, $B_c = \tilde{B}_w - \tilde{B}_u M$, $C_c = \tilde{C} - \tilde{D}_u K$, and $D_c = \tilde{D}_w - \tilde{D}_u M$. Note that in this case $\tilde{D}_u$ and $\tilde{D}_w$ are zero matrices.

System



Performance



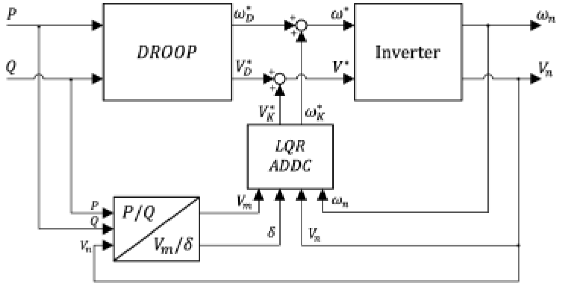


Data-driven control (Trickett, PMAPS 2024)

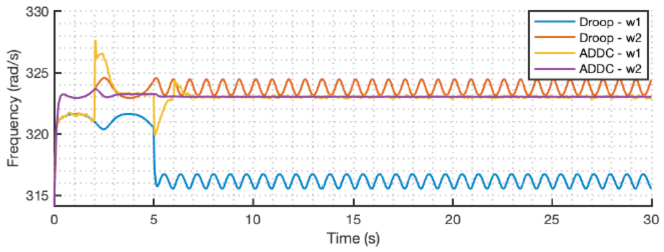
Formulation

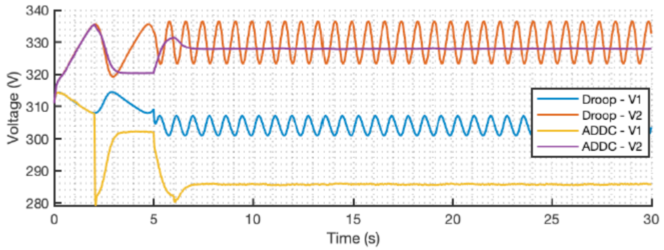
$$\begin{cases} \begin{bmatrix} -P + X_1 M X_1^\top + \mu^2 RVR^\top + \frac{1}{\eta_1} I & -X_1 M \\ -MX_1^\top & M - V \end{bmatrix} \preceq 0 \\ L - U_0 Q P^{-1} Q^\top U_0^\top \geq 0 \\ P \geq I \\ X_0 Q = P \\ \operatorname{trace}(P) \quad + \operatorname{trace}(L) + \operatorname{trace}(V) \leq \gamma \end{cases}$$

System



Performance





Inverse optimal and passivity-based control (Hallinan, IEEE CDC 2023)

Formulation

Lemma 3: Consider system (6) with linear node and edge dynamics as in (22). For each $i \in \mathcal{V}$ with dynamics (1), consider matrices $Y_i \in \mathbb{R}^{n\times n}$, $S_i \in \mathbb{R}^{m\times m}$ and $\Gamma_i \in \mathbb{R}^{n\times n}$ that satisfy the following set of linear matrix inequalities (LMIs):

$$\begin{bmatrix} Y_i A_i^T + A_i Y_i + 2B_{ui} S_i B_{ui}^T & Y_i & B_i - Y_i C_i^T \\ Y_i & -\Gamma_i & 0 \\ B_i^T - C_i Y_i & 0 & 0 \end{bmatrix} \le 0 \tag{23a}$$

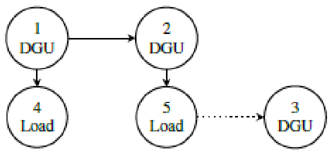
$$-\tfrac{1}{2} B_{ui} S_i B_{ui}^T - \tfrac{1}{2}\left(A_i Y_i + Y_i A_i^T\right) > 0 \tag{23b}$$

$$Y_i = Y_i^T > 0, \quad \Gamma_i > 0, \quad S_i = S_i^T > 0 \tag{23c}$$

Considering $P_i = Y_i^{-1}$ and $R_i = -\frac{1}{2} S_i^{-1}$ yields the following distributed local controller for each $i \in \mathcal{V}$:

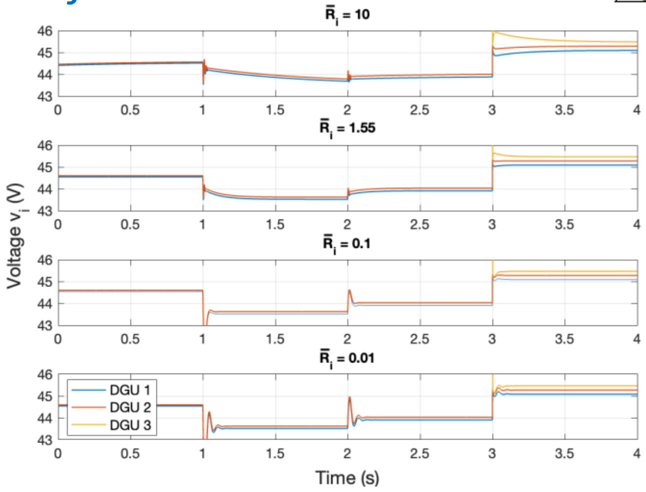
$$u_i^* = -\tfrac{1}{2} R_i^{-1} B_{ui}^T P_i x_i \tag{24}$$

System



Converter Parameters			
r_i	$0.2\,\Omega$	v_1^{set}	48 V
c_i	2.2 mF	v_2^{set}	47.8 V
l_i	1.8 mH	v_3^{set}	48.1 V
g_i	$\frac{1}{100}$ S	z_i	$1\,\Omega$
Load Parameters			
g_4	0.1 S	g_5	0.05 S
c_4	70 μF	c_5	70 μF
Line Parameters			
r_{ij}	$0.05\,\Omega$	l_{ij}	2.1 μH

Performance



Unbalanced optimal power-flow

- Back in 2017, optimal power-flow for unbalanced networks was mostly unexplored.
- Published in IEEE Trans. Sustainable Energy.

Formulation

$$\text{minimize} \tag{28}$$

$$f(c) = [[\boldsymbol{M}[\boldsymbol{Y}]^{-1}\boldsymbol{I_{inj}}] * [\boldsymbol{M}[\boldsymbol{Y}]^{-1}\boldsymbol{I_{inj}}]]^{\boldsymbol{T}} \cdot \boldsymbol{R} + \sum_{x=1}^{s}(1-\eta_x)\left(\sqrt{c_{x,real}{}^2 + c_{x,reactive}{}^2}\right) \sum_{i \in N} \frac{v_i \cdot v_i^*}{R_{N,i}}$$

with respect to:

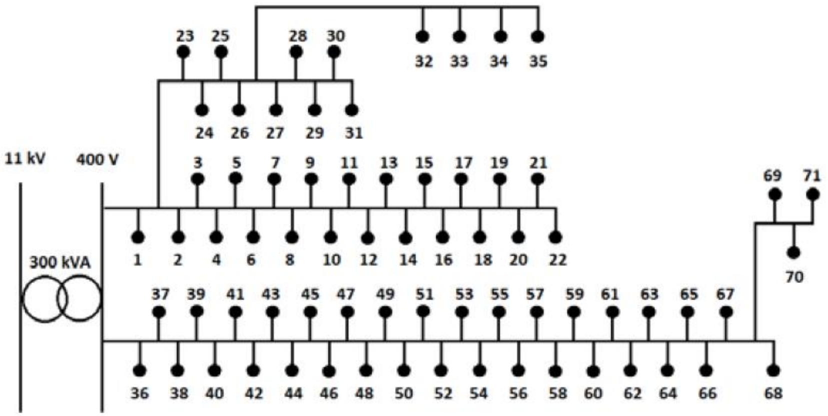
$c_{x,real}, c_{x,reactive}$ for all ESSs x

subject to

$$\left.\begin{array}{lr}
g_{1a}(\boldsymbol{c}) = \left(\sum_{x,\text{such that } \phi(x)\in A} c_{x,real}\right) - c_{tp,a} = 0 & (28.1)\\
g_{1b}(\boldsymbol{c}) = \left(\sum_{x,\text{such that } \phi(x)\in B} c_{x,real}\right) - c_{tp,b} = 0 & (28.2)\\
g_{1c}(\boldsymbol{c}) = \left(\sum_{x,\text{such that } \phi(x)\in C} c_{x,real}\right) - c_{tp,c} = 0 & (28.3)\\
|I_{ij}|^2 \le \overline{I_{ij}}^2 \text{ for all branches connecting nodes } (i,j) & (28.4)\\
v_{i,real}{}^2 + v_{i,imag}{}^2 \le \overline{V_i}^2, \text{for all } i, i \notin N & (28.5)\\
-K_{1A}v_{i,real} - K_{2A}v_{i,imag} \le -\underline{V_i}\ i \in A & (28.6)\\
-K_{1B}v_{i,real} - K_{2B}v_{i,imag} \le -\underline{V_i}\ i \in B & (28.7)\\
-K_{1C}v_{i,real} - K_{2C}v_{i,imag} \le -\underline{V_i}\ i \in C & (28.8)\\
VUF_m{}^2 \le \overline{VUF_m}^2, \text{for all buses } m & (28.9)\\
v_{i,real}{}^2 + v_{i,imag}{}^2 \le \overline{V_{neutral}}^2, i \in N & (28.10)\\
c_{x,real}{}^2 + c_{x,reactive}{}^2 \le \overline{c_x}^2 \text{ for all } ESSs\ x & (28.11)\\
-c_{x,real} \le \frac{E_{capacity,x}}{\Delta t}\left(SoC_x - \overline{SoC_x}\right) & (28.12)\\
c_{x,real} \le \frac{E_{capacity,x}}{\Delta t}\left(SoC_x - \underline{SoC_x}\right) & (28.13)
\end{array}\right\}$$

Convexified constraints

Case study



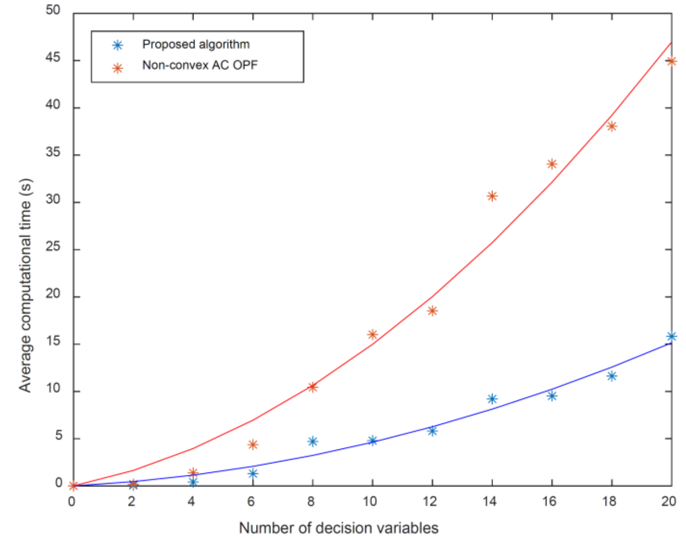
Small approximation error

TABLE II. OPTIMALITY STATISTICS IN THE CASE STUDIES

Case	Network	ESS efficiency variance	Maximum optimality gap	Average optimality gap
1	Radial	0%	0.07%	0.01%
2	Radial	2%	0.35%	0.03%
3	Radial	4%	0.93%	0.15%
4	Meshed	0%	0.02%	0.00%

Results

Less computational expense



22% reduction in energy losses